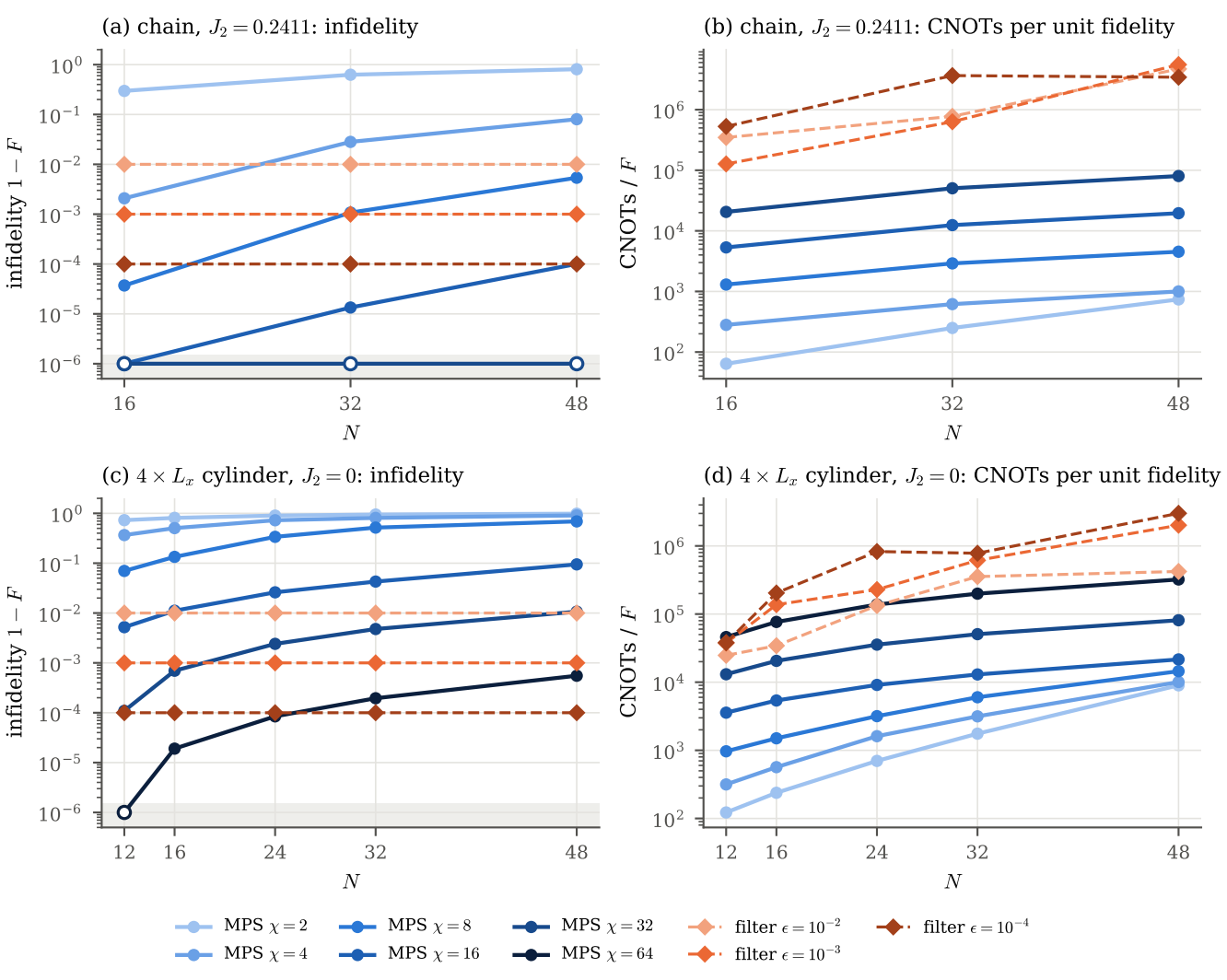
(a) chain, $J_2 = 0.2411$: infidelity
infidelity $1 - F$
$10^{0}$
$10^{-1}$
$10^{-2}$
$10^{-3}$
$10^{-4}$
$10^{-5}$
$10^{-6}$
16
32
48
$N$
(b) chain, $J_2 = 0.2411$: CNOTs per unit fidelity
CNOTs / $F$
$10^{6}$
$10^{5}$
$10^{4}$
$10^{3}$
$10^{2}$
16
32
48
$N$
(c) $4 \times L_x$ cylinder, $J_2 = 0$: infidelity
infidelity $1 - F$
$10^{0}$
$10^{-1}$
$10^{-2}$
$10^{-3}$
$10^{-4}$
$10^{-5}$
$10^{-6}$
12
16
24
32
48
$N$
(d) $4 \times L_x$ cylinder, $J_2 = 0$: CNOTs per unit fidelity
CNOTs / $F$
$10^{6}$
$10^{5}$
$10^{4}$
$10^{3}$
$10^{2}$
12
16
24
32
48
$N$
MPS $\chi = 2$
MPS $\chi = 4$
MPS $\chi = 8$
MPS $\chi = 16$
MPS $\chi = 32$
MPS $\chi = 64$
filter $\epsilon = 10^{-2}$
filter $\epsilon = 10^{-3}$
filter $\epsilon = 10^{-4}$